

Thunderstorms

Thunderstorms are rain showers that produce powerful electrical discharges called lightning

Thunderstorms come in different shapes and sizes. Single-cell storms typically last 20–30 minutes and may produce heavy rainfall and hail. Larger multicell storms may generate weak tornadoes. The most severe thunderstorms are supercells, which are bigger and may produce violent tornadoes. Severe thunderstorms are classed as those with hailstones 1 in (2.5 cm) in diameter, winds of more than 56 mph (90 kph), or tornadoes.

Lightning strikes

As tiny water droplets and ice crystals in the cold tops of the clouds bump into each other to form bigger drops of rain and hailstones, these particles develop positive and negative electrical charges. It is this difference in charge (potential difference) between the particles that creates lightning, as a flow of electrons moves from the negatively charged particles to the positive. Lightning is extremely hot, and as it passes through air it causes the air to expand instantaneously. This expansion creates a shockwave—the familiar rumble of thunder. Lightning zigzags between one part of the cloud and another, between the cloud and air, or between the cloud and the ground.

BLUE JETS

This type of lightning is found 25–30 miles (40–50 km) above the ground, projecting from the top of a large storm cell to the upper atmosphere.



Cumulonimbus clouds

The lower part is composed of water droplets but the upper regions are composed of supercooled droplets and ice crystals.

back-sheared anvil

rear downdraft

interior cloud wall

Cyclone core

This is the area of the storm cell where warm air spirals upward through the cloud.

flanking line, a mass of developing cumulus clouds extending from the main storm

circulating banks of wall cloud

► SUPERCELL

No two supercell storms are the same but they do all share most of the features shown here: a huge cumulonimbus cloud, wind blowing around an area of low atmospheric pressure, a large updraft of warm air, heavy rainfall and hail, and downdrafts of cooled air.

Rain-free side

Rain does not fall from every part of the cloud.

Warm air inflow

Rising air in the middle of the cell produces low pressure near the ground. This sucks in more warm air and keeps the system going.

rotating updraft

Tornado

Destructive, rapidly spinning vortices may develop beneath about 30 percent of large supercells.

the rapid vortex of a tornado whips up soil and debris from the ground

Cloud-to-ground lightning

These powerful discharges reach a temperature of 54,000 °F (30,000 °C).